

Static vs Contextual Embeddings for Lexical Ambiguity

WiC evaluation in eight steps

Choudhary Prashant Santosh

Enrollment no. 1910474
University of Trier MSc Natural Language Processing

Rahul Khunt

Enrollment no. 1911272
Trends in Natural Language Processing

1 Introduction

- **Lexical ambiguity** means that words such as **bank** have more than one meaning.
- Static GloVe uses one vector for every occurrence [2].
- Contextual BERT changes the vector with the sentence [3].
- **Aim:** test which representation captures sense differences better.

2 Definitions

- **Homonymy:** unrelated meanings. **Polysemy:** related meanings [6].
- **Embedding:** a word represented as a numerical vector.
- **Cosine similarity:** how closely two vector directions align.
- **WiC task:** does the same word have the same meaning in two sentences? [1]

3 Related work

- WordNet and dictionary-based word sense disambiguation.
- Multi-sense embeddings with several vectors per word [4].
- Contextual models such as ELMo and BERT.
- WiC as a context-sensitive benchmark, introduced in 2019 [1].

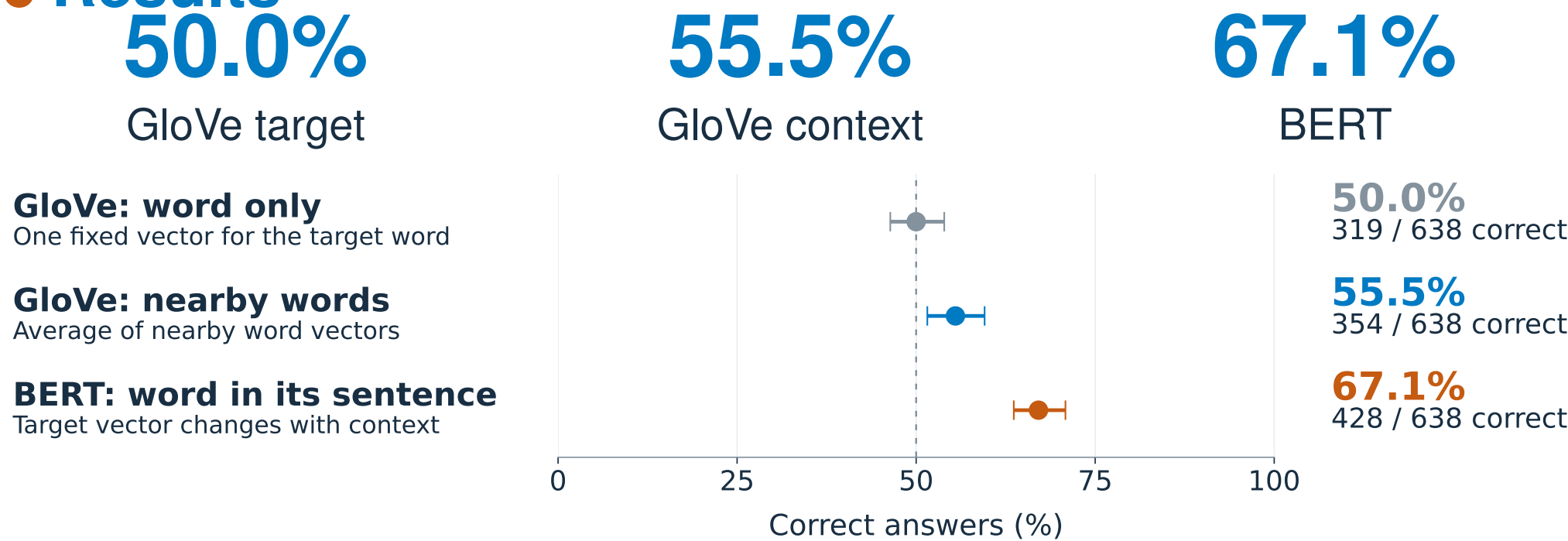
4 Hypotheses

- H1: the static target diagnostic remains at chance.
- H2: nearby GloVe context gives a small improvement.
- H3: frozen BERT beats GloVe context without fine-tuning.

5 Methodology

- WiC: 5,428 train, 638 validation and 1,400 public test pairs without public labels.
- Systems: GloVe target, GloVe context ± 2 , BERT last-four-layer mean.
- Cosine threshold and settings selected on training data only.
- Validation used once. 1,000 paired bootstrap resamples.

6 Results



BERT gains 11.6 points over GloVe context. Paired 95% interval: 6.6 to 16.9. BERT macro F1 is 66.7% and ROC-AUC is 0.716. Layer 9 reaches 66.6%, while layer 12 reaches 63.6%.

7 Error analysis

- BERT right and GloVe wrong: 178. GloVe right and BERT wrong: 104.
- BERT same-sense recall: 77.4%. Different-sense recall: 56.7%.
- GloVe context same-sense recall: 46.4%. Different-sense recall: 64.6%.
- BERT errors: 210 of 638.

8 Conclusion

BERT captures sense differences better than GloVe on this frozen WiC protocol. It still reaches only 67.1%. The evidence comes from one small English validation set, one frozen BERT model and a simple GloVe baseline.

References

[1] Pilehvar & Camacho-Collados (2019). *WiC*. NAACL.
[2] Pennington, Socher & Manning (2014). *GloVe*. EMNLP.

[3] Devlin et al. (2019). *BERT*. NAACL.
[4] Camacho-Collados & Pilehvar (2018). *From Word to Sense Embeddings*. JAIR.

[5] Loureiro et al. (2021). *Language Models for Word Sense Disambiguation*. Computational Linguistics.
[6] Haber & Poesio (2021). *Polysemy and Homonymy in Contextualised Language Models*. Findings of EMNLP.